
(a)
AdamW
Muon
step
500
1000
2000
4000
6000
10000
20000
d0
d1
d2
1.65
1.50
1.55
1.66
1.43
1.42
1.54
1.64
2.26
2.02
2.01
2.28
2.29
(b)
AdamW vs Muon
AdamW, other data order
Muon, other data order
JS (bits)
step
(c)
JS (bits)
training loss
(d)
AdamW
Muon
validation loss
step